

PROBLEM-SOLUTION FIT VALIDATION

Validating Educational Pains and Technical Capabilities Alignment

1. Problem Statement

Traditional written tests fail to capture a student's active recall and verbal communication capabilities. Verbal testing (vivas/presentations) is ideal, but manually grading hours of audio explanations is highly subjective, labor-intensive, and unfeasible for large classes. EdTech tools currently lack local, private, and automatic systems for speech-concept analysis.

2. Proposed Solution Fit

The Voice-Based Concept Understanding Analyser (VBCUA) solves this bottleneck by establishing an automated pipelines:

- **Speech to Text**: Bypasses manual transcription by converting spoken audio to text locally using Whisper.
- **Semantic Scoring**: Matches the candidate's explanation against a verified reference concept using Sentence-BERT embeddings, checking for conceptual understanding rather than mere keyword match.
- **Fluency Assessment**: Detects filler words and silent hesitations, scoring vocal projection (average RMS energy) to grade speaking confidence.

3. Verification of Fit

The fit is verified when the automatic grading system provides scores that map to standard human evaluations (Strong, Moderate, Poor), flags specific filler word habits, and outputs downloadable PDFs that students can use to guide self-study.